

# Directional Analysis of Space Radiation in Low Earth Orbit Using the Timepix3 Detector Onboard OneWeb JoeySat

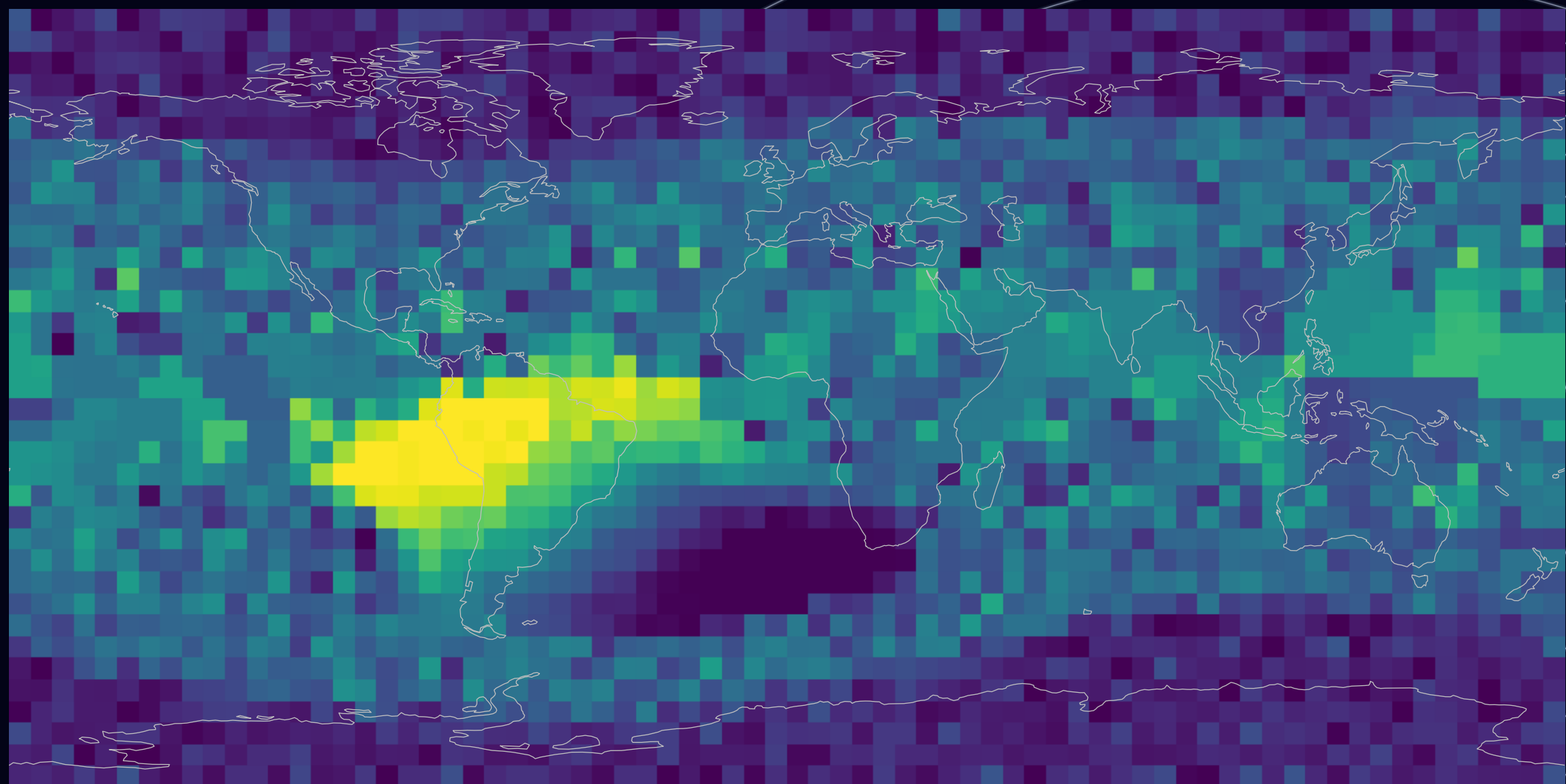
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Phase I, 613 km. Median elevation angle  $\beta$  of protons per 5° cell. Inside the anomaly  $\beta$  follows the magnetic inclination: high near the magnetic equator, low towards the south. Outside it the gradient is an order of magnitude weaker.

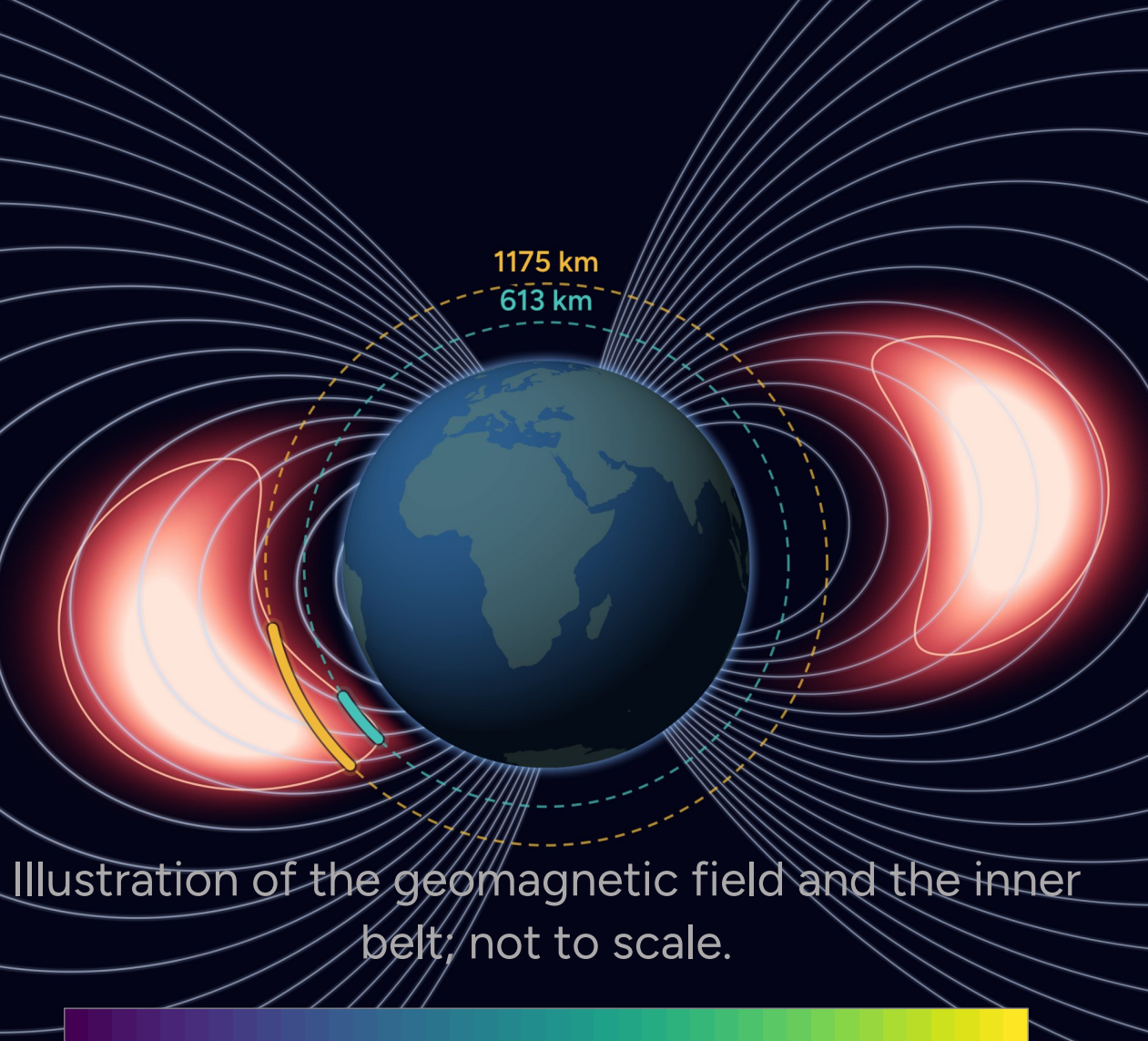
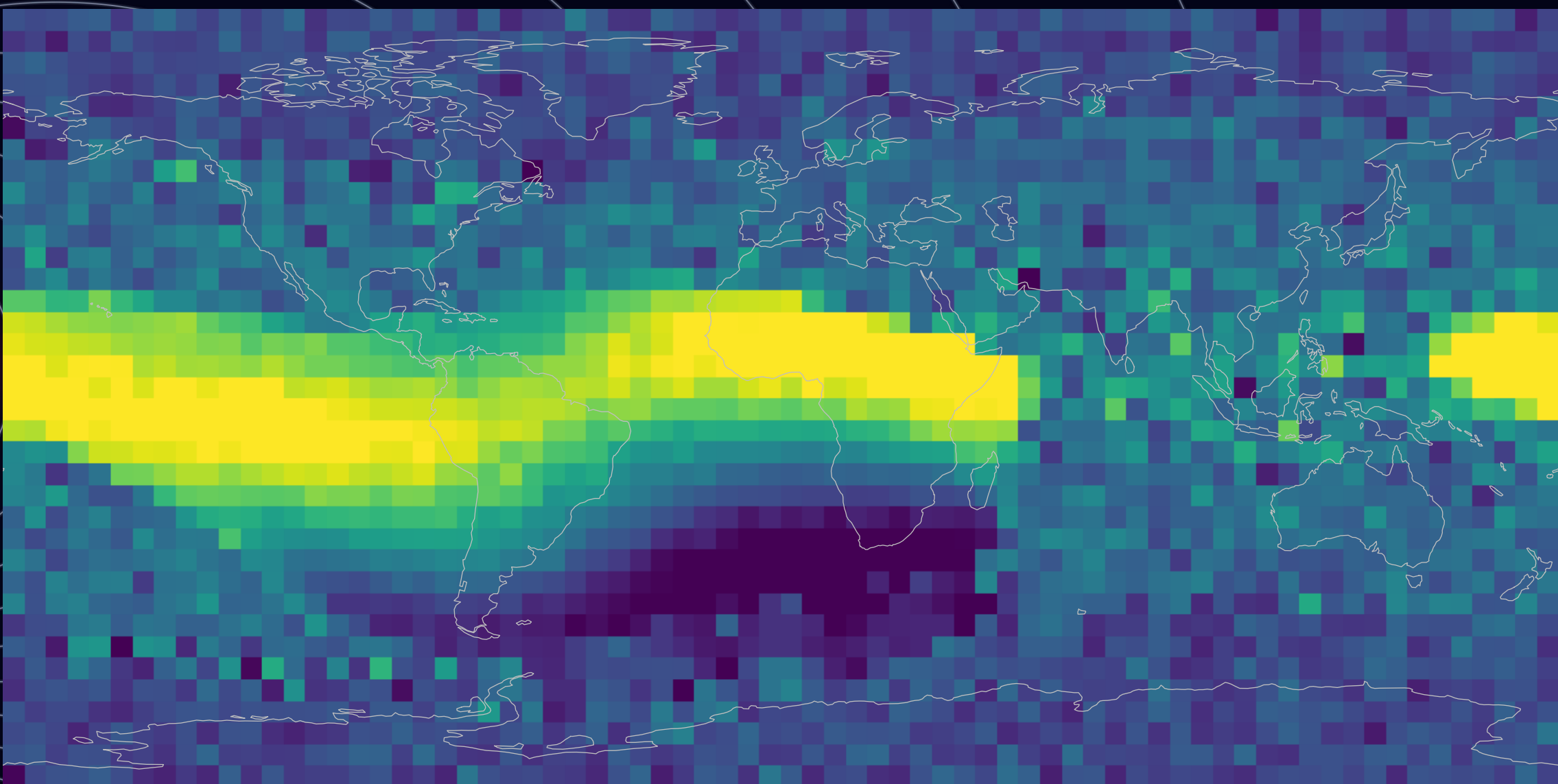


Illustration of the geomagnetic field and the inner belt, not to scale.  
median elevation angle  $\beta$  [deg]  
one scale for both maps



Phase II, 1,175 km. The same quantity at the higher altitude, where the anomaly spans 242° of longitude. The gradient across it has the same sense, from the same detector, analysed identically.

A single 500  $\mu\text{m}$  silicon layer read out by a Timepix3 chip, without a coincidence telescope, resolves that the arrival directions of protons inside the South Atlantic Anomaly are organized by the local magnetic field.

1 layer

500  $\mu\text{m}$  of silicon, no coincidence telescope

2 orbits

613 km and 1,175 km, analysed identically

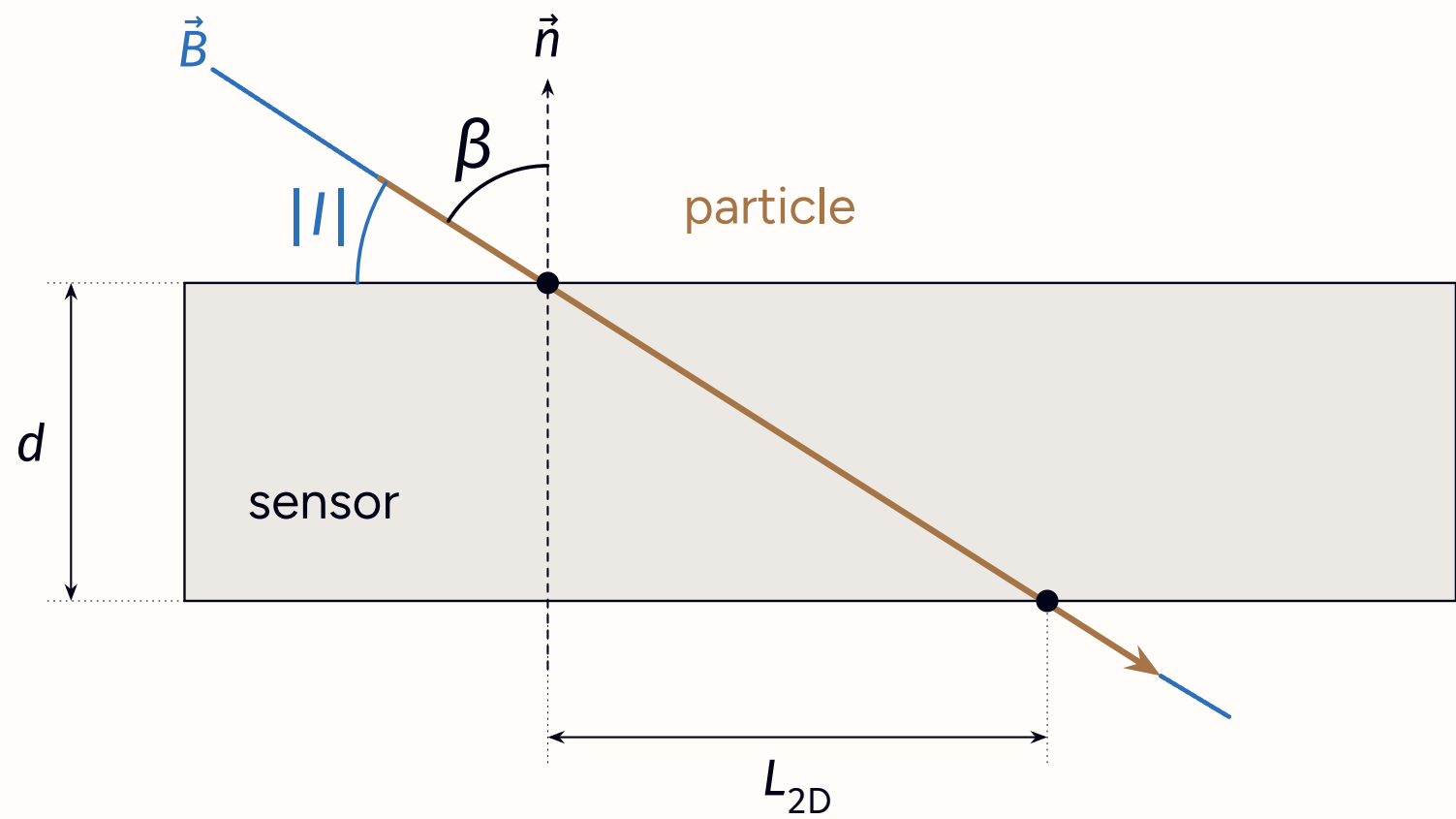
9,265

protons in the core of the anomaly

180×

measured slope over the noise of the isotropic control

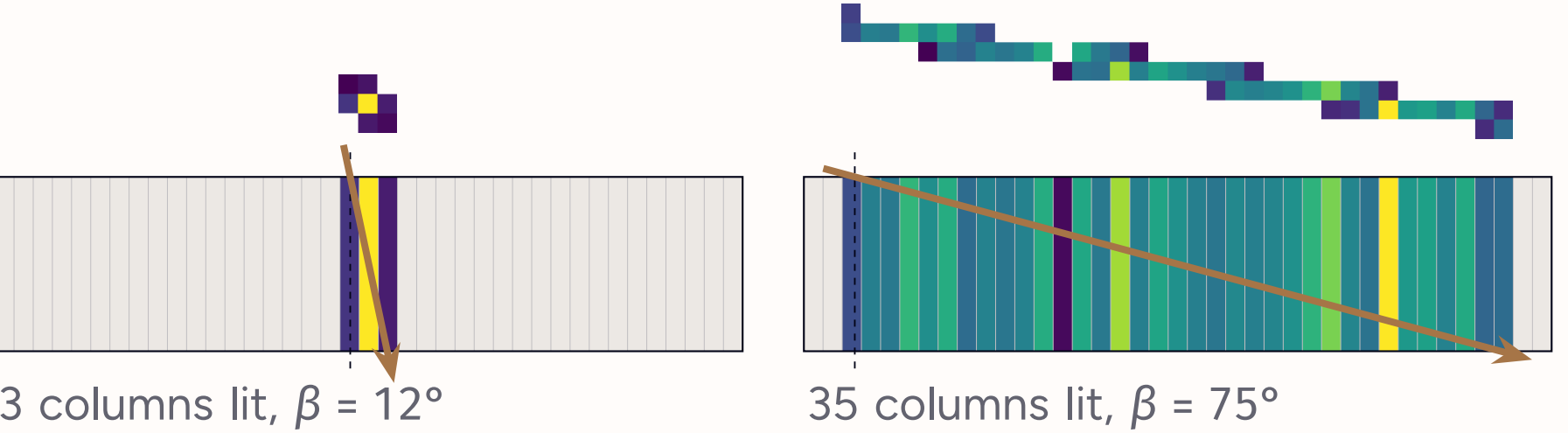
## Method: the elevation angle



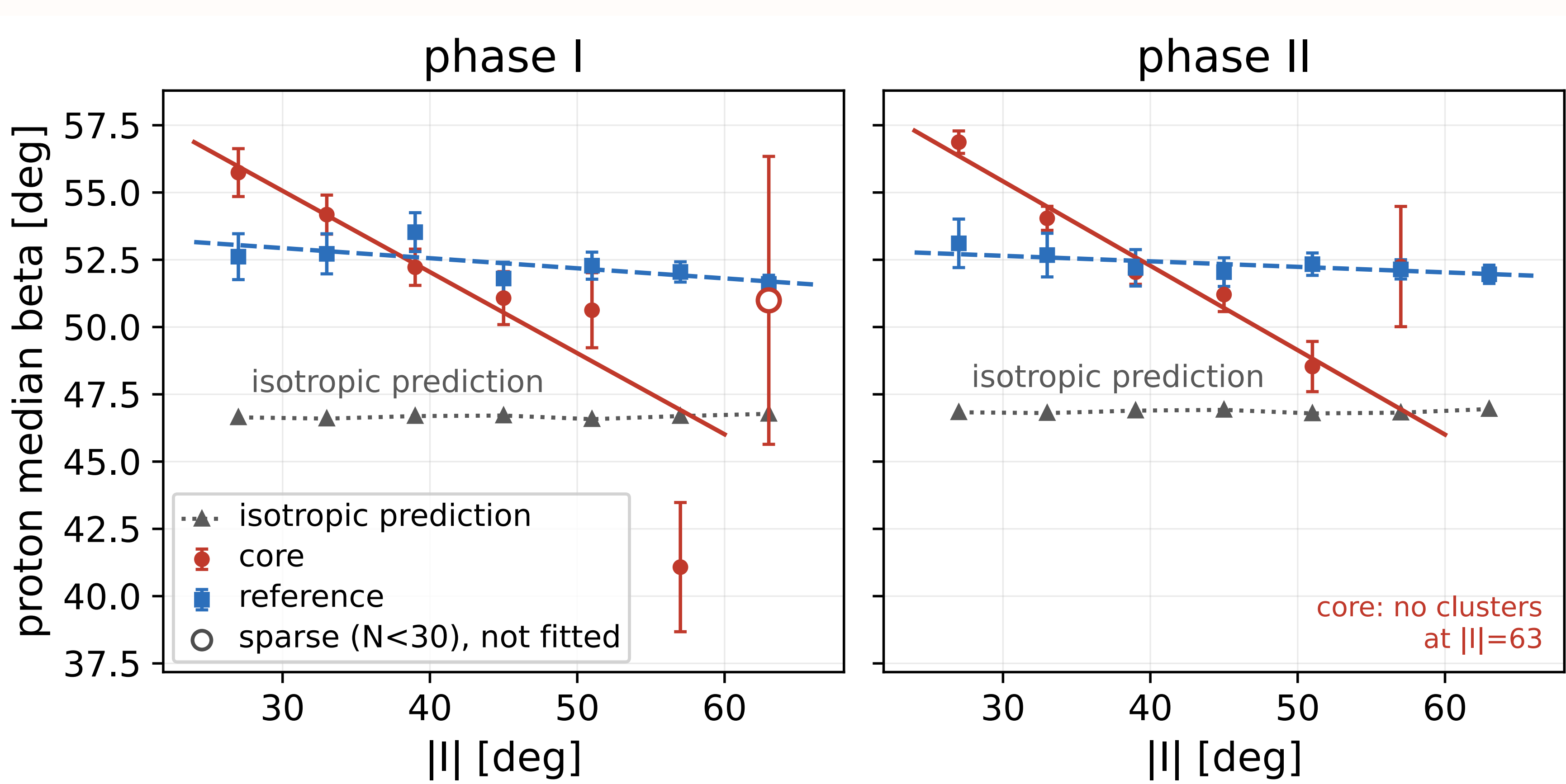
$$\beta = \arctan\left(\frac{p L_{2D}}{d}\right)$$

- > **One layer:**  $\beta$  follows from the projected length of the pixel cluster.
- > **Charge sharing** makes the cluster wider than the track itself; that excess is subtracted before  $\beta$  is computed.
- > **Floor and ceiling:** charge spread sets 11° for protons, the read-out area 85.9°.
- > **Selection:** only clusters narrower than 1.8 pixels enter the comparison, the limit at which the width still measures a single track.

Two measured clusters, each drawn above the sensor columns that record it. Both cross sections and both clusters are at the same pixel scale, so the cluster is visibly wider than the track.



## Result: median elevation angle against magnetic inclination



**Core:** at least 10° inside the anomaly contour; **reference:** at least 10° outside it; triangles: isotropic control; lines: weighted fits.

- > **Observable:** the median  $\beta$  in bins of local magnetic inclination  $|I|$ . A field-aligned population would follow  $\beta = 90^\circ - |I|$ ; an isotropic one gives a flat profile.
- > **Inside the anomaly** the median  $\beta$  falls with the local magnetic inclination, in both phases; outside it the profile is an order of magnitude shallower. The same inclination occurs in both regions, which makes them comparable.
- > **Null:** an isotropic population passed through the identical reconstruction chain gives a **flat** profile (sampling noise 0.0017 per degree); the measured slopes exceed that noise by a factor of **180**.
- > **Sign, not magnitude:** negative under every selection, control variable, spectral band and time interval, while a placebo anomaly drawn in the reference shows none. The magnitude,  $-0.62$  to  $-0.12$  per degree, depends on the cluster-width limit, so no calibrated slope is quoted.

## Instrument: MiniPIX Space



- > **Timepix3** reading 500  $\mu\text{m}$  of silicon, 128 × 128 pixels of 55  $\mu\text{m}$ , behind 5 mm of aluminium.
- > **Upper deck,** sensor to the zenith; 25 s frames every 64 s.

## Limitations

- > **Magnitude:** no plateau in the width or guard-band scan, so no calibrated slope.
- > **Pitch angles:** not interpreted here; that needs the shield in the angular response.
- > **Species:** heavy ions excluded by a floor of 53° to 58°; electrons scattered beyond 45° in the shield.
- > **Sense of travel:** an axis, not a vector; the east–west asymmetry cannot be tested.
- > **Altitudes:** no quantitative contrast between the two phases is claimed.

## Outlook: the geomagnetic field

- > **Cause and effect in one data set:** the proton directions follow the inclination of the geomagnetic field given by IGRF. The detector therefore records the radiation and, through its directions, the imprint of the field that organizes it, with no magnetometer on board.
- > **A changing field:** the field is driven by the geodynamo and the anomaly drifts with its secular variation. Directional data of this kind **could follow how the trapped population responds to changes of the field**, including short-term ones.
- > **Coverage:** a single instrument samples the anomaly for minutes per orbit. **More detectors of this size across a constellation** would cover the field continuously and make such tracking possible.

Within the limits set by the lateral spread of the collected charge and by the shield, the method recovers directional structure of the trapped environment from an instrument that fits a small satellite.



Interactive version: vary the cluster-width limit, the guard band and the occupancy threshold and watch the maps, profiles and slopes respond.

ADVASPACE  
radiation data

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